

Project Title: Design and Implementation of a Small Language Model using Sparse Mixture of Experts (top-2 Gating)

Project Objective. The goal of this project is to design, implement, and evaluate a multimodal (language-logic-maths-coding) transformer model based on a Sparse Mixture of Experts (MoE) architecture using OLMoE, an open-source framework. The model must process and integrate natural language, programming code, logic, and math reasoning tasks using selected datasets.

The final deliverables include:

1. A fully functional and runnable source code hosted on GitHub (with documentation).
2. A detailed 15-page technical report explaining architecture, training, datasets, metrics, graphs, ablation studies, and experimental results.
3. Models and logs sufficient to replicate results.
4. **Use of generative AI (e.g., ChatGPT, GitHub Copilot, etc.) is allowed for code and report generation, provided the final output is fully working and verifiable.**

Core Specifications

1. Model Architecture

- Base model: Vision-Language Transformer using Sparse Mixture of Experts
- Framework: [OLMoE](#) (Open Language Mixture of Experts)
- Expert Selection: Top-2 gating (each token routed to top 2 experts)
- Number of experts: 8–16 (adjustable during experimentation)
- Expert implementation: MLP with SwiGLU activation
- Precision: bfloat16 throughout training and inference
- Positional Encoding: Rotary or learnable (experiment and report findings)
- Cross-attention for vision-language fusion

2. NLP Dataset Selection

Choose one/two dataset(s) from the following:

mC4 (multilingual C4) Clean Corpus from Common Crawl

OSCAR Multilingual web corpus, with clean content for NLP

CC100 Web data from Wikipedia/CCNet

Wikipedia dumps Encyclopedic texts in each language

Multimodal Text Datasets

Modality	Dataset(s)
Code	HuggingFace CodeParrot, The Stack v2 (Python)
Logic	LogiQA, ReClor
Mathematics	MATH, GSM8k, Minerva
Education	SAT, GRE, GMAT, ENS (French ENS exam)

Tasks must be tokenized and batched in a multi-task setup. Include mixture-of-tasks routing logic and design decisions.

Implementation Plan

Tools & Libraries

- Python ≥ 3.10
- PyTorch (or JAX)
- HuggingFace Transformers and Datasets
- Accelerate / DeepSpeed (for memory efficiency)
- OLMoE (fork or extend the existing repo)
- wandb or TensorBoard for logging

GitHub Repository Requirements

- README.md: Project summary, setup instructions, and usage
- requirements.txt or environment.yml
- configs/: JSON or YAML model configuration files
- scripts/: Training, evaluation, logging scripts
- notebooks/: Exploratory code and visualizations
- data/: Scripts or pointers for dataset preparation
- src/: Model and training code
- report/: Final PDF and supporting LaTeX/Jupyter files

Make the repository easy to replicate and runnable from a clean environment.

Evaluation Metrics

Each task must be evaluated with relevant metrics:

Task Type	Metrics
VQA	Accuracy, VQA-Score
Programming	Pass@k, CodeBLEU
Logic	Accuracy
Math Reasoning	Exact match, Step-by-step Accuracy
University Exams	Score %, normalized scores
MoE	Expert load, Expert entropy, Routing loss

Deliverables

1. GitHub Repo with
 - Working model & scripts
 - Fully reproducible results
 - README with detailed usage
2. Technical Report (PDF + source files)
3. Logs of experiments (WandB/Tensorboard or local JSON)
4. Final checkpoints and sample outputs